

## 2021 考研数学试卷答案速查 (数学一)

### 一、选择题

- (1) (D)      (2) (C)      (3) (A)      (4) (B)      (5) (B)  
(6) (A)      (7) (C)      (8) (D)      (9) (C)      (10) (B)

### 二、填空题

- (11)  $\frac{\pi}{4}$       (12)  $\frac{2}{3}$       (13)  $y = x^2$   
(14)  $4\pi$       (15)  $\frac{3}{2}$       (16)  $\frac{1}{5}$

### 三、解答题

$$(17) \text{ 原式} = \lim_{x \rightarrow 0} \frac{\left(1 + \int_0^x e^{t^2} dt\right) \sin x - e^x + 1}{(e^x - 1) \sin x} \quad (2 \text{ 分})$$

$$= \lim_{x \rightarrow 0} \frac{\sin x - e^x + 1}{(e^x - 1) \sin x} + \lim_{x \rightarrow 0} \frac{\sin x \int_0^x e^{t^2} dt}{(e^x - 1) \sin x} \quad (4 \text{ 分})$$

$$= \lim_{x \rightarrow 0} \frac{x + o(x^2) + 1 - \left[1 + x + \frac{1}{2}x^2 + o(x^2)\right]}{x^2} + \lim_{x \rightarrow 0} \frac{\int_0^x e^{t^2} dt}{x} \quad (7 \text{ 分})$$

$$= \lim_{x \rightarrow 0} \frac{-\frac{1}{2}x^2 + o(x^2)}{x^2} + \lim_{x \rightarrow 0} e^{x^2} \quad (9 \text{ 分})$$

$$= \frac{1}{2} \quad (10 \text{ 分})$$

$$(18) \quad u_n(x) = e^{-nx} + \frac{x^{n+1}}{n(n+1)} \quad (n=1, 2, \cdots)$$

$$(1) \quad \text{设 } v_n(x) = e^{-nx}, \quad w_n(x) = \frac{x^{n+1}}{n(n+1)}, \quad \text{则 } u_n(x) = v_n(x) + w_n(x)$$

$$\sum_{n=1}^{\infty} e^{-nx} \text{ 收敛区间为 } (0, +\infty), \quad \sum_{n=1}^{\infty} \frac{x^{n+1}}{n(n+1)} \text{ 收敛区间为 } (-1, 1) \quad (3 \text{ 分})$$

$$x=0 \text{ 时, } \sum_{n=1}^{\infty} u_n(x) = \sum_{n=1}^{\infty} 1, \quad \text{级数发散}$$

$$x=1 \text{ 时, } \sum_{n=1}^{\infty} \frac{1}{n(n+1)}, \quad \text{级数收敛}$$

所以级数的收敛域为  $(0, 1]$ . (4 分)

$$(2) \quad S_1(x) = \sum_{n=1}^{\infty} v_n(x) = \sum_{n=1}^{\infty} e^{-nx} = \lim_{n \rightarrow \infty} \frac{e^{-x}[1-e^{-nx}]}{1-e^{-x}} = \frac{e^{-x}}{1-e^{-x}} (x > 0) \quad (6 \text{ 分})$$

$$S_2(x) = \sum_{n=1}^{\infty} w_n(x) = \sum_{n=1}^{\infty} \frac{x^{n+1}}{n(n+1)}$$

$$\text{则 } S_2''(x) = \sum_{n=1}^{\infty} x^{n-1} = \frac{1}{1-x} (x \neq 1)$$

$$S_2'(x) - S_2'(0) = \int_0^x \frac{1}{1-t} dt = -\ln|1-x|, \quad \text{因为 } S_2'(0) = 0, \quad \text{所以 } S_2'(x) = -\ln|1-x| (x \neq 1)$$

$$S_2(x) - S_2(0) = \int_0^x -\ln|1-t| dt = (1-x)\ln|1-x| + x, \quad \text{因为 } S_2(0) = 0, \quad \text{所以}$$

$$S_2(x) = (1-x)\ln|1-x| + x (x \neq 1) \quad (9 \text{ 分})$$

$$\text{因此 } x \neq 1 \text{ 时, } S(x) = S_1(x) + S_2(x) = \frac{e^{-x}}{1-e^{-x}} + (1-x)\ln|1-x| + x$$

$$\text{当 } x = 1 \text{ 时, 和函数连续, 所以 } S(1) = \lim_{x \rightarrow 1} \left( \frac{e^{-x}}{1-e^{-x}} + (1-x)\ln|1-x| + x \right) = \frac{e^{-1}}{1-e^{-1}} + 1$$

$$\text{所以, } S(x) = \begin{cases} \frac{e^{-x}}{1-e^{-x}} + (1-x)\ln|1-x| + x, & 0 < x < 1 \\ \frac{e^{-1}}{1-e^{-1}} + 1 & x = 1 \end{cases} \quad (12 \text{ 分})$$

(19) 根据题意, 目标函数为  $f(x, y, z) = z^2$ , 约束条件是  $x^2 + 2y^2 - z - 6 = 0$  以及  $4x + 2y + z - 30 = 0$  (2 分)

$$\text{设 } F(x, y, z, \lambda, \mu) = z^2 + \lambda(x^2 + 2y^2 - z - 6) + \mu(4x + 2y + z - 30)$$

$$\begin{cases} F'_x = 2\lambda x + 4\mu = 0 \\ F'_y = 4\lambda y + 2\mu = 0 \\ F'_z = 2z - \lambda + \mu = 0 \\ F'_\lambda = x^2 + 2y^2 - z - 6 = 0 \\ F'_\mu = 4x + 2y + z - 30 = 0 \end{cases} \quad (6 \text{ 分})$$

解得  $(x, y, z) = (4, 1, 12)$  或者  $(x, y, z) = (-8, -2, 66)$  (10 分)

$$f(4, 1, 12) = 12^2, \quad f(-8, -2, 66) = 66^2$$

因此距离的最大值为 66 (12 分)

(20)

(1) 根据题意, 易知  $D_1 = \{x^2 + y^2 \leq 4\}$

$$\iint_{D_1} (4 - x^2 - y^2) dx dy = \int_0^{2\pi} \int_0^2 (4 - r^2) r dr d\theta = 8\pi \quad (4 \text{ 分})$$

$$(2) \int_{\partial D_1} \frac{(xe^{x^2+4y^2} + y)dx + (4ye^{x^2+4y^2} - x)dy}{x^2 + 4y^2}$$

$$P(x, y) = \frac{xe^{x^2+4y^2} + y}{x^2 + 4y^2}, Q(x, y) = \frac{4ye^{x^2+4y^2} - x}{x^2 + 4y^2}$$

$$\frac{\partial P(x, y)}{\partial y} = \frac{8xye^{x^2+4y^2}(x^2 + 4y^2 - 1) + x^2 - 4y^2}{(x^2 + 4y^2)^2} = \frac{\partial Q(x, y)}{\partial x}$$

补充曲线  $l_1: x^2 + 4y^2 = \varepsilon^2$  (顺时针方向)

$$\begin{aligned} & \int_{\partial D_1} \frac{(xe^{x^2+4y^2} + y)dx + (4ye^{x^2+4y^2} - x)dy}{x^2 + 4y^2} \\ &= \oint_{\partial D_1 + l_1} \frac{(xe^{x^2+4y^2} + y)dx + (4ye^{x^2+4y^2} - x)dy}{x^2 + 4y^2} - \oint_{l_1} \frac{(xe^{x^2+4y^2} + y)dx + (4ye^{x^2+4y^2} - x)dy}{x^2 + 4y^2} \end{aligned}$$

由格林公式可知,

$$\oint_{\partial D_1 + l_1} \frac{(xe^{x^2+4y^2} + y)dx + (4ye^{x^2+4y^2} - x)dy}{x^2 + 4y^2} = \iint_{D_2} \left( \frac{\partial Q(x, y)}{\partial x} - \frac{\partial P(x, y)}{\partial y} \right) dx dy = 0$$

其中  $D_2$  为  $\partial D_1$  和  $l_1$  围成的封闭区域. (8 分)

$$\oint_{l_1} \frac{(xe^{x^2+4y^2} + y)dx + (4ye^{x^2+4y^2} - x)dy}{x^2 + 4y^2} = \frac{1}{\varepsilon^2} \oint_{l_1} (xe^{x^2+4y^2} + y)dx + (4ye^{x^2+4y^2} - x)dy$$

根据格林公式

$$\frac{1}{\varepsilon^2} \oint_{l_1} (xe^{x^2+4y^2} + y)dx + (4ye^{x^2+4y^2} - x)dy = -\frac{1}{\varepsilon^2} \iint_{D_3} -2 dx dy = \frac{1}{\varepsilon^2} \cdot 2\pi \cdot \varepsilon \cdot \frac{1}{2} \varepsilon = \pi$$

其中  $D_3$  是  $l_1$  围成的封闭区域.

$$\text{所以 } \int_{\partial D_1} \frac{(xe^{x^2+4y^2} + y)dx + (4ye^{x^2+4y^2} - x)dy}{x^2 + 4y^2} = -\pi \quad (12 \text{ 分})$$

(21)

$$(1) \quad |\lambda E - A| = \begin{vmatrix} \lambda - a & -1 & 1 \\ -1 & \lambda - a & 1 \\ 1 & 1 & \lambda - a \end{vmatrix} = (\lambda - a + 1)^2 (\lambda - a - 2)$$

令  $|\lambda E - A| = 0$ , 解得  $\lambda_1 = \lambda_2 = a - 1, \lambda_3 = a + 2$  (2 分)

$$(a - 1)E - A \rightarrow \begin{pmatrix} 1 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \text{ 解得 } a_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, a_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

$$(a + 2)E - A \rightarrow \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \text{ 解得 } a_3 = \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix} \quad (4 \text{ 分})$$

将  $a_1, a_2$  进行施密特正交化可得  $\beta_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \beta_2 = \frac{(a_2, \beta_1)}{(\beta, \beta_1)} \beta_1 = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$  (6 分)

将  $(\beta_1, \beta_2, a_3)$  单位化, 可得  $\gamma_1 = \begin{pmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{pmatrix}, \gamma_2 = \begin{pmatrix} \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \end{pmatrix}, \gamma_3 = \begin{pmatrix} -\frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{pmatrix},$

可得正交矩阵  $P = \begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{pmatrix}$ , 使  $P^T A P = \Lambda = \begin{pmatrix} a - 1 & 0 & 0 \\ 0 & a - 1 & 0 \\ 0 & 0 & a + 2 \end{pmatrix}$  (8 分)

(2) 因为  $P^T A P = \Lambda$  可知  $A = P \Lambda P^T$ ,

$$C^2 = (a + 3)E - A = P[(a + 3)E - \Lambda]P^T = P \begin{pmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{pmatrix} P^T = P \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} P^T P \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} P^T$$

因为  $C$  为正定矩阵, 所以

$$C = P \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} P^T = \begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{pmatrix}^T$$

$$= \begin{pmatrix} \frac{5}{3} & -\frac{1}{3} & \frac{1}{3} \\ -\frac{1}{3} & \frac{5}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{5}{3} \end{pmatrix} \quad (12 \text{ 分})$$

(22)

易知  $X + Y = 2, Y > X > 0$ , 且  $X$  在  $(0, 1)$  上服从均匀分布;

$$(I) \quad X \text{ 的概率密度 } f_X(x) = \begin{cases} 1, & 0 < x < 1 \\ 0, & \text{其他} \end{cases}. \quad (4 \text{ 分})$$

$$(II) \quad Z \text{ 的分布函数 } F_Z(z) = P\{Z \leq z\} = P\left\{\frac{2-X}{X} \leq z\right\};$$

$$z < 1 \text{ 时, } F_Z(z) = 0; z \geq 1 \text{ 时, } F_Z(z) = \int_{\frac{2}{z+1}}^1 1 dx = 1 - \frac{2}{z+1};$$

$$Z \text{ 的概率密度为 } f_Z(z) = \begin{cases} \frac{2}{(z+1)^2}, & z \geq 1 \\ 0, & \text{其他} \end{cases}. \quad (8 \text{ 分})$$

$$(III) \quad E\left(\frac{X}{Y}\right) = E\left(\frac{X}{2-X}\right) = \int_0^1 \frac{x}{2-x} dx = \int_0^1 \left(\frac{2}{2-x} - 1\right) dx$$

$$= -2 \ln |2-x| \Big|_0^1 - 1 = 2 \ln 2 - 1 \quad (12 \text{ 分})$$